

CEH Exam Questions (№ 0.1)

Question 1

The attacker tries to find the servers of the attacked company. He uses the following command: ***nmap 192.168.1.64/28***

The scan was successful, but he didn't get any results. Identify why the attacker could not find the server based on the following information:

The attacked company used network address 192.168.1.64 with mask 255.255.255.192. In the network, the servers are in the addresses 192.168.1.122, 192.168.1.123 and 192.168.1.124.

- a. **He is scanning from 192.168.1.64 to 192.168.1.78 because of the mask /28 and the servers are not in that range**
- b. He needs to change the address to 192.168.1.0 with the same mask
- c. The network must be down and the nmap command and IP address are ok
- d. He needs to add the command ""ip address"" just before the IP address

Question 2

Which of the following best describes the operation of the Address Resolution Protocol?

- a. It sends a reply packet for a specific IP, asking for the MAC address
- b. It sends a request packet to all the network elements, asking for the domain name from a specific IP
- c. It sends a reply packet to all the network elements, asking for the MAC address from a specific IP
- d. **It sends a request packet to all the network elements, asking for the MAC address from a specific IP**

Question 3

John needs to send a super-secret message, and for this, he wants to use the technique of hiding a secret message within an ordinary message. The technique provides "security through obscurity." Which of the following techniques will John use?

- a. **Steganography**

- b. Deniable encryption
- c. Encryption
- d. Digital watermarking

Question 4

How can resist an attack using rainbow tables?

- a. **Use password salting**
- b. Use of non-dictionary words
- c. Lockout accounts under brute force password cracking attempts
- d. All uppercase character passwords

Question 5

What flags will be set when scanning when using the following command:

```
# nmap -sX host.companydomain.com
```

- a. SYN flag is set
- b. **URG, PUSH and FIN are set**
- c. SYN and ACK flags are set
- d. ACK flag is set

Question 6

In what type of testing does the tester have some information about the internal work of the application?

- a. **Grey-box**
- b. White-box
- c. Announced
- d. Black-box

Question 7

An attacker tries to infect as many devices connected to the Internet with malware as possible to get the opportunity to use their computing power and functionality for

automated attacks hidden from the owners of these devices. Which of the proposed approaches fits description of the attacker's actions?

- a. APT attack
- b. **Creating a botnet**
- c. Mass distribution of Ransomware
- d. Using Banking Trojans

Question 8

After scanning the ports on the target machine, you see a list of open ports, which seems unusual to you:

1. Starting NMAP 5.21 at 2019-06-18 12:32
2. NMAP scan report for 172.19.40.112
3. Host is up (1.00s latency).
4. Not shown: 993 closed ports
5. PORT STATE SERVICE
6. 21/tcp open ftp
7. 23/tcp open telnet
8. 80/tcp open http
9. 139/tcp open netbios-ssn
10. 515/tcp open
11. 631/tcp open ipp
12. 9100/tcp open
13. MAC Address: 00:00:5D:3F:EE:92

Based on the NMAP output, identify what is most likely this host?

- a. The host is likely a Windows machine
- b. **The host is likely a printer**
- c. The host is likely a Linux machine
- d. The host is likely a router

Question 9

Which of the following is a vulnerability in modern processors such as Intel, AMD and ARM using speculative execution?

- a. **Spectre and Meltdown**
- b. Launch Daemon
- c. Application Shimming
- d. Named Pipe Impersonation

Question 10

Identify the attack by the description:

It is the wireless version of the phishing scam. This is an attack-type for a rogue Wi-Fi access point that appears to be a legitimate one offered on the premises but has been set up to eavesdrop on wireless communications. When performing this attack, an attacker fools wireless users into connecting a device to a tainted hotspot by posing as a legitimate provider. This type of attack may be used to steal the passwords of unsuspecting users by either snooping the communication link or by phishing, which involves setting up a fraudulent website and luring people there.

- a. **Evil Twin**
- b. Collision
- c. Signal Jamming
- d. Sinkhole

Question 11

Due to the network slowdown, the IT department decided to monitor the Internet traffic of all employees to track a possible cause, but they can't do it immediately. Which of the following is troublesome to take this kind of measure from a legal point of view?

- a. All of the employees would stop normal work activities.
- b. **Not informing the employees that they are going to be monitored could be an invasion of privacy.**
- c. The absence of an official responsible for traffic on the network
- d. Lack of comfortable working conditions

Question 12

You are working in a cybersecurity company and notice an unusual spike in network traffic from a particular IP address. You deploy an AI-powered intrusion detection system (IDS) to analyze the traffic. The AI flags the IP address for potential malicious activity due to abnormal behavior patterns. What should be your next step in leveraging AI to mitigate this potential attack?

- a. Deploy an AI bot to hack back the source
- b. Manually block the IP address
- c. Use AI to further analyze past traffic patterns from this IP
- d. Ignore the alert as AI often generates false positives

Question 13

What is an example of an AI-powered offensive technique in hacking?

- a. AI intrusion detection systems
- b. Automated malware generation**
- c. Behavioral analysis in IDS
- d. AI-driven encryption cracking

Question 14

Which AI method is most effective in detecting Advanced Persistent Threats (APTs)?

- a. Neural networks
- b. Deep learning**
- c. Reinforcement learning
- d. Clustering

Question 15

Evil Russian hacker Ivan is attacking again! This time, he got a job in a large American company to steal commercial information for his customer to gain a competitive advantage in the market. In his attack, Ivan used all available means, especially blackmail, bribery, and technological surveillance. What is the name of such an attack?

- a. Social Engineering

- b. Information Leakage
- c. Corporate Espionage**
- d. Business Loss

Question 16

What is the name of the risk assessment method that allows you to study how various types of negative events (violations, failures or destructions) can affect the main activities of the company and key business processes?

- a. Risk Mitigation
- b. Emergency Plan Response (EPR)
- c. Business Impact Analysis (BIA)**
- d. Disaster Recovery Planning (DRP)

Question 17

Which mode of a NIC (interface) allows you to intercept and read each network packet that arrives in its entirety?

- a. Multicast
- b. Simplex Mode
- c. Promiscuous mode**
- d. Port forwarding

Question 18

Which of the following types of keys does the Heartbleed bug expose to the Internet, making exploiting any compromised system very easy?

- a. Root
- b. Public
- c. Shared
- d. Private**

Question 19

Jenny, a pentester, conducts events to detect viruses in systems. She uses a detection method where the anti-virus executes the malicious codes on a virtual machine to simulate CPU and memory activities. Which of the following methods does Jenny use?

- a. Heuristic Analysis
- b. Integrity checking
- c. Code Emulation**
- d. Vulnerability scanner

Question 20

IPsec is a suite of protocols developed to ensure the integrity, confidentiality, and authentication of data communications over an IP network. Which protocol is NOT included in the IPsec suite?

- a. Security Association (SA)
- b. Authentication Header (AH)
- c. Media Access Control (MAC)**
- d. Encapsulating Security Protocol (ESP)

Question 21

What Linux command will you use to resolve a domain name into an IP address?

- a. host -t soa resolveddomain.com
- b. host -t a resolveddomain.com**
- c. host -t ns resolveddomain.com
- d. host -t AXFR resolveddomain.com

Question 22

As a result of the attack on the dating web service, Ivan received a dump of all user passwords in a hashed form. Ivan recognized the hashing algorithm and started identifying passwords. What tool is he most likely going to use if the service used hashing without salt?

- a. Brute force
- b. XSS
- c. Dictionary attacks

d. **Rainbow table**

Question 23

Which of the following modes of IPSec should you use to assure integrity and confidentiality of data within the same LAN?

- a. **ESP transport mode**
- b. AH transport mode
- c. ESP tunnel mode
- d. AH tunnel mode

Question 24

Lisandro is engaged in sending spam. To avoid blocking, he connects to incorrectly configured SMTP servers that allow e-mail relay without authentication (which allows Lisandro to fake information about the sender's identity). What is the name of such an SMTP server?

- a. Public SMTP server.
- b. **Open mail relay**
- c. Message transfer agent
- d. Weak SMTP

Question 25

How is AI used in mitigating Distributed Denial of Service (DDoS) attacks?

- a. By deploying bots to attack back
- b. By simulating DDoS attacks for training
- c. By rerouting all traffic through AI networks
- d. By predicting and blocking abnormal traffic patterns

Question 26

The analyst needs to evaluate the possible threats to Blackberry phones for third-party company. To do this, he will use the Blackjacking attack method to demonstrate how an

attacker could circumvent perimeter defences and gain access to the corporate network. Which of the following tools is best suited for the analyst for this task?

- a. Paros Proxy
- b. BBCrack
- c. Blooover
- d. **BBProxy**

Question 27

What is the term for a machine learning model that can generate realistic, human-like text?

- a. **Generative Adversarial Network (GAN)**
- b. Long Short-Term Memory (LSTM)
- c. Natural Language Processing (NLP)
- d. Recurrent Neural Network (RNN)

Question 28

Which of the following is the type of message that sends the client to the server to begin a 3-way handshake while establishing a TCP connection?

- a. RST
- b. **SYN**
- c. ACK
- d. SYN-ACK

Question 29

A healthcare company is under cyber attack, where hackers are trying to exploit vulnerabilities in its IoT medical devices. You have deployed an AI-based defense system that automatically detects anomalies in device communications and sends alerts when unauthorized access is detected. How can AI improve the defense of IoT devices in this scenario?

- a. **AI can predict and stop potential IoT exploits before they occur**
- b. AI will isolate all IoT devices on the network
- c. AI will physically block unauthorized devices from connecting
- d. AI will deploy new patches to all IoT devices instantly

Question 30

What is the primary advantage of using AI for threat detection in cybersecurity?

- a. AI can automatically remove malware
- b. Higher speed in scanning networks
- c. **Higher accuracy compared to traditional methods**
- d. AI requires no human supervision

Question 31

Identify the type of DNS configuration in which first DNS server on the internal network and second DNS in DMZ?

- a. EDNS
- b. DynDNS
- c. **Split DNS**
- d. DNSSEC

Question 32

Which of the following is most useful for quickly checking for SQL injection vulnerability by sending a special character to web applications?

- a. Backslash
- b. **Single quotation**
- c. Semicolon
- d. Double quotation

Question 33

Rajesh, a black-hat hacker, could not find vulnerabilities in the target company's network since their infrastructure is very well protected. IDS, firewall with strict rules, etc. He is trying to find such an attack method independent of the reliability of the infrastructure of this company. Which attack is an option suitable for Rajesh?

- a. Buffer Overflow
- b. **Social Engineering**

- c. Denial-of-Service
- d. Confidence trick

Question 34

Identify the type of partial breaks in which the attacker discovers a functionally equivalent algorithm for encryption and decryption, but without learning the key?

- a. **Global deduction**
- b. Information deduction
- c. Instance deduction
- d. Total break

Question 35

During the security audit, Gabriella used Wget to read exposed information from a remote server and got this result:

```
Server: nginx/1.21.0
Date: Mon, 02 Aug 2021 13:29:13 EST
Content-Type: text/html
Content-Length: 5683
Last-Modified: Thu, 05 Jul 2021 17:44:09 EST
Connection: keep-alive
ETag: "5bb65169-1633"
Accept-Ranges: bytes
```

What is the name of this method of obtaining information?

- a. SQL injection
- b. XML External Entities (XXE)
- c. Cross-site scripting
- d. **Banner grabbing**

Question 36

The Domain Name System (DNS) is the phonebook of the Internet. When a user tries to access a web address like “example.com”, web browser or application performs a DNS Query against a DNS server, supplying the hostname. The DNS server takes the hostname and resolves it into a numeric IP address, which the web browser can connect to. Which of the proposed tools allows you to set different DNS query types and poll arbitrarily specified servers?

- a. Nikto
- b. Metasploit
- c. **Nslookup**
- d. Wireshark

Question 37

Jack needs to analyze the files produced by several packet-capture programs such as Wireshark, tcpdump, EtherPeek and WinDump. Which of the following tools will Jack use?

- a. **Tcptrace**
- b. Tcptracroute
- c. Nessus
- d. OpenVAS

Question 38

Which machine learning technique is commonly used to identify anomalies in network traffic?

- a. Reinforcement learning
- b. Supervised learning
- c. **Unsupervised learning**
- d. Transfer learning

Question 39

John, a cybersecurity specialist, wants to perform a syn scan in his company's network. He has two machines. The first machine (192.168.0.98) has snort installed, and the second machine (192.168.0.151) has kiwi Syslog installed. When he started a syn scan in the network, he notices that kiwi Syslog is not receiving the alert message from snort. He

decides to run Wireshark in the snort machine to check if the messages are going to the kiwi Syslog machine. What Wireshark filter will show the connections from the snort machine to kiwi Syslog machine?

- a. **tcp.dstport==514 && ip.dst==192.168.0.151**
- b. tcp.dstport==514 && ip.dst==192.168.0.0/16
- c. tcp.srcport==514 && ip.src==192.168.0.98
- d. tcp.srcport==514 && ip.src==192.168.151

Question 40

Assume an attacker gained access to the internal network of a small company and launches a successful STP manipulation attack. What are his next steps?

- a. He will repeat this action so that it escalates to a DoS attack
- b. He will repeat the same attack against all L2 switches of the network
- c. **He will create a SPAN entry on the spoofed root bridge and redirect traffic to his computer**
- d. He will activate OSPF on the spoofed root bridge

Question 41

Which of the following is an attack where used precomputed tables of hashed passwords?

- a. Brute Force Attack
- b. **Rainbow Table Attack**
- c. Hybrid Attack
- d. Dictionary Attack

Question 42

Which of the following best describes of counter-based authentication system?

- a. An authentication system that bases authentication decisions on physical attributes
- b. **An authentication system that creates one-time passwords that are encrypted with secret keys**
- c. An authentication system that bases authentication decisions on behavioural attributes

- d. An authentication system that uses passphrases that are converted into virtual passwords

Question 43

Identify which term corresponds to the following description:

It is can potentially adversely impact a system through unauthorized access, destruction, disclosure, denial of service or modification of data.

- a. Attack
- b. **Threat**
- c. Vulnerability
- d. Risk

Question 44

An attacker gained access to a Linux host and stolen the password file from /etc/passwd. Which of the following scenarios best describes what an attacker can do with this file?

- a. Nothing because he cannot read the file because it is encrypted
- b. **Nothing because the password file does not contain the passwords themselves**
- c. The attacker can perform actions as root because the file reveals the passwords to the root user only
- d. The attacker can perform actions as a user because he can open it and read the user ids and corresponding passwords

Question 45

Alex works as a network administrator at ClassicUniversity. There are many Ethernet ports are available for professors and authorized visitors (but not for students) on the university campus. However, Alex realized that some students connect their notebooks to the wired network to have Internet access. He identified this when the IDS alerted for malware activities in the network. What should Alex do to avoid this problem?

- a. Ask students to use the wireless network
- b. Separate students in a different VLAN
- c. Disable unused ports in the switches
- d. **Use the 802.1x protocol**

Question 46

Alex, a cybersecurity science student, needs to fill in the information into a secured PDF-file job application received from a prospective employer. He can't enter the information because all the fields are blocked. He doesn't want to request a new document that allows the forms to be completed and decides to write a script that pulls passwords from a list of commonly used passwords to try against the secured PDF until the correct password is found or the list is exhausted. Which attack is the student attempting?

- a. Brute-force attack
- b. **Dictionary-attack**
- c. Session hijacking
- d. Man-in-the-middle attack

Question 47

How can AI be used to detect and prevent malware attacks?

- a. **All of the above**
- b. By examining file signatures
- c. By analyzing network traffic patterns
- d. By identifying suspicious behavior

Question 48

Which type of AI attack involves manipulating input data to deceive a machine learning model into making incorrect predictions?

- a. Poisoning attack
- b. Adversarial attack
- c. Denial-of-service attack
- d. **Evasion attack**

Question 49

What of the following is the most common method of using "ShellShock" or "Bash Bug"?

- a. Manipulate format strings in text fields

- b. **Through Web servers utilizing CGI to send a malformed environment variable**
- c. Using SSH
- d. Using SYN Flood

Question 50

Which of the following is correct?

- a. **Sniffers operate on Layer 2 of the OSI model**
- b. Sniffers operate on both Layer 2 & Layer 3 of the OSI model
- c. Sniffers operate on Layer 4 of the OSI model
- d. Sniffers operate on Layer 3 of the OSI model

Question 51

What property is provided by using hash?

- a. Availability
- b. Confidentiality
- c. **Integrity**
- d. Authentication

Question 52

Maria, the leader of the Blue Team, wants to use network traffic analysis to implement the ability to detect an intrusion in her network of several hosts quickly. Which tool is best suited to perform this task?

- a. **NIDS**
- b. Honeypot
- c. HIDS
- d. Firewalls

Question 53

Sniffing is a process of monitoring and capturing all data packets passing through a given network. An intruder can capture and analyze all network traffic by placing a packet sniffer

on a network in promiscuous mode. Sniffing can be either Active or Passive in nature. How does passive sniffing work?

- a. This is the process of sniffing through the router
- b. This is the process of sniffing through the switch
- c. **This is the process of sniffing through the hub**
- d. This is the process of sniffing through the gateway

Question 54

A large enterprise is experiencing sporadic network outages, and the security team suspects that it might be a Distributed Denial of Service (DDoS) attack. They decide to use AI-driven tools to detect and mitigate the attack. The AI system starts analyzing the incoming network traffic and flags an unusually high number of requests from a specific geographic region. However, you notice that legitimate users are also located in the same region. How can the security team balance blocking the DDoS traffic without disrupting legitimate users?

- a. Slow down the network traffic for all users to handle the load
- b. Ignore the flagged traffic since legitimate users are present
- c. **Deploy a CAPTCHA system for all users in the flagged region**
- d. Completely block all traffic from the flagged region

Question 55

Which of the following is a potential risk associated with using AI in cybersecurity?

- a. Increased privacy concerns
- b. Ethical implications
- c. Overreliance on AI systems
- d. **All of the above**

Question 56

To send an email using SMTP protocol which does not encrypt messages and leaving the information vulnerable to being read by an unauthorized person. To solve this problem, SMTP can upgrade a connection between two mail servers to use TLS, and the transmitted

emails will be encrypted. Which of the following commands is used by SMTP to transmit email over TLS?

- a. FORCETLS
- b. **STARTTLS**
- c. UPGRADETLS
- d. OPPORTUNISTICTLS

Question 57

Leonardo, an employee of a cybersecurity firm, conducts an audit for a third-party company. First of all, he plans to run a scanning that looks for common misconfigurations and outdated software versions. Which of the following tools is most likely to be used by Leonardo?

- a. Armitage
- b. **Nikto**
- c. Nmap
- d. Metasploit

Question 58

Black-hat hacker Ivan wants to determine the status of ports on a remote host. He wants to do this quickly but imperceptibly for IDS systems. For this, he uses a half-open scan that doesn't complete the TCP three-way handshake. What kind of scanning does Ivan use?

- a. XMAS scans
- b. PSH Scan
- c. **TCP SYN (Stealth) Scan**
- d. FIN scan

Question 59

Organizations need to deploy a web-based software package that requires three separate servers and internet access. What is the recommended architecture in terms of server placement?

- a. All three servers need to be placed internally

- b. All three servers need to face the Internet so that they can communicate between themselves
- c. A web server and the database server facing the Internet, an application server on the internal network
- d. **A web server facing the Internet, an application server on the internal network, a database server on the internal network**

Question 60

What is the primary purpose of adversarial machine learning in cybersecurity?

- a. To improve the accuracy of AI models
- b. To enhance the efficiency of AI algorithms
- c. To generate synthetic data for training AI models
- d. **To detect and prevent attacks on AI systems**

Question 61

Your company's threat intelligence team uses AI to conduct automated penetration testing on a new cloud infrastructure. During the test, the AI identifies several misconfigurations that could lead to data exposure. These findings were discovered much faster than with manual testing. What is the most likely advantage of using AI in this penetration testing scenario?

- a. AI can fix misconfigurations immediately
- b. **AI identifies complex patterns in misconfigurations that manual testing may miss**
- c. AI can only detect vulnerabilities after a breach occurs
- d. AI replaces human penetration testers completely

Question 62

Identify a tool that can be used for passive OS fingerprinting?

- a. nmap
- b. tracert
- c. ping
- d. **tcpdump**

Question 63

While performing online banking using a browser, your friend receives a message that contains a link to a website. He decides to click on this link, and another browser session starts and displays a funny video. A few hours later, he receives a letter from the bank stating that his online bank was visited from another country and tried to transfer money. The bank also asks him to contact them and confirm the transfer if he really made it. What vulnerability did the attacker use when attacking your friend?

- a. **Cross-Site Request Forgery**
- b. Clickjacking
- c. Webform input validation
- d. Cross-Site Scripting

Question 64

Which AI tool can be used for automating the process of discovering security misconfigurations?

- a. Anomaly detection systems
- b. Honeypots
- c. Supervised classifiers
- d. **AI-driven bots**

Question 65

What type of cryptography is used in IKE, SSL, and PGP?

- a. Hash
- b. Secret Key
- c. **Public Key**
- d. Digest

Question 66

Which of the following is a component of IPsec that performs protocol-level functions required to encrypt and decrypt the packets?

- a. Oakley
- b. Internet Key Exchange (IKE)
- c. **IPsec driver**
- d. IPsec Policy Agent

Question 67

What is the term for a machine learning algorithm that can learn from data without being explicitly programmed?

- a. Reinforcement learning
- b. Deep learning
- c. Supervised learning
- d. **Unsupervised learning**

Question 68

How can AI be used to improve the efficiency of incident response teams?

- a. By providing real-time threat intelligence
- b. By recommending mitigation strategies
- c. By automating incident triage
- d. **All of the above**

Question 69

The ping utility is used to check the integrity and quality of connections in networks. In the process, it sends an ICMP Echo-Request and captures the incoming ICMP Echo-Reply, but quite often remote nodes block or ignore ICMP. Which of the options will solve this problem?

- a. Use arping
- b. Use traceroute
- c. **Use hping**
- d. Use broadcast ping

Question 70

Monitoring your company's assets is one of the most important jobs you can perform. What warnings should you try to reduce when configuring security tools, such as security information and event management (SIEM) solutions or intrusion detection systems (IDS)?

- a. Only False Positives
- b. **False Positives and False Negatives**
- c. Only True Negatives
- d. True Positives and True Negatives

Question 71

To protect the enterprise infrastructure from the constant attacks of the evil hacker Ivan, Viktor divided the network into two parts using the network segmentation approach.

- In the first one (local, without direct Internet access), he isolated business-critical resources.
- In the second (external, with Internet access), he placed public web servers to provide services to clients.

Subnets communicate with each other through a gateway protected by a firewall. What is the name of the external subnet?

- a. **Demilitarized Zone**
- b. Network access control
- c. WAF
- d. Bastion host

Question 72

The CIA Triad is a security model that highlights the main goals of data security and serves as a guide for organizations to protect their confidential data from unauthorized access and data theft. What are the three concepts of the CIA triad?

- a. Transference, transformation and transcendence
- b. Comparison, reflection and abstraction
- c. **Confidentiality, integrity, and availability**
- d. Efficiency, equity and liberty

Question 73

Lisandro is a novice fraudster, he uses special software purchased in the depths of the network for sending his malware. This program allows it to deceive pattern-based detection mechanisms and even some behavior-based ones, disguising malwares as harmless programs. What does Lisandro use?

- a. Ransomware
- b. Dropper
- c. Payload
- d. **Crypter**

Question 74

An e-commerce platform implements AI to continuously monitor user accounts for abnormal login attempts. Over a week, the system notices a rise in login attempts from unfamiliar locations and devices. The AI system flags these as potential brute-force attacks. How could AI help mitigate this threat in real time?

- a. Sending a warning email to administrators without taking action
- b. Manually reviewing each flagged attempt
- c. Slowing down the entire network to prevent further attacks
- d. **Automatically blocking the suspicious IP addresses**

Question 75

You need to conduct a technical assessment of the network for a small company that supplies medical services. All computers in the company use Windows OS. What is the best approach for discovering vulnerabilities?

- a. Create a disk image of a clean Windows installation
- b. Use the built-in Windows Update tool
- c. Check MITRE.org for the latest list of CVE findings
- d. **Use a scan tool like Nessus**

Question 76

Which of the following nmap options can be used for very fast scanning

- a. -O
- b. -T0
- c. -T4
- d. **-T5**

Question 77

Implementing the security testing process early in the SDLC is the key to finding out and fixing the security bugs early in the SDLC lifecycle. The security testing process can be performed in two ways, Automated or Manual web application security testing. Which of the proposed statements is true?

- a. **Automatic and manual testing should be used together to better cover potential problems**
- b. Neural networks and artificial intelligence are already used in new tools and do not require additional actions
- c. Automatic testing requires a lot of money and is still very imperfect, so it cannot be used for security
- d. Manual testing is obsolete and should be completely replaced by automatic testing

Question 78

Identify the way to achieve chip-level security of an IoT device?

- a. **Encrypting the JTAG interface**
- b. Changing the password of the router
- c. Turning off the device when not needed or not in use
- d. Closing insecure network services

Question 79

Which of the following is an entity in a PKI that will vouch for the identity of an individual or company?

- a. VA
- b. **CA**
- c. KDC
- d. CR

Question 80

Your team uses AI-based anomaly detection software that identifies patterns of cyber attacks in real-time. Recently, it detected a new type of malware that attempts to encrypt files and hold them for ransom. The AI system was able to stop the malware before any files were encrypted. What AI method is likely being used to detect this ransomware?

- a. Rule-based detection
- b. Signature-based detection
- c. **Behavior-based anomaly detection**
- d. Static file analysis

Question 81

What is the first and most important phase that is the starting point for penetration testing in the work of an ethical hacker?

- a. Scanning
- b. Maintaining Access
- c. Gaining Access
- d. **Reconnaissance**

Question 82

A rootkit is a clandestine computer program designed to provide continued privileged access to a computer while actively hiding its presence. They are classified according to the place of their injection.

What type of rootkit loads itself underneath the computer's operating system and can intercept hardware calls made by the original operating system.

- a. Application rootkit
- b. **Hypervisor (Virtualized) Rootkits**
- c. Kernel mode rootkits
- d. Memory rootkit

Question 83

The absolute majority of routers and switches use packet filtering firewalls. That kind of firewalls makes decisions about allowing traffic to pass into the network based on the information contained in the packet header. At what level of the OSI model do these firewalls work?

- a. Physical layer
- b. Session layer
- c. Network layer**
- d. Application layer

Question 84

The flexible SNMP architecture allows you to monitor and manage all network devices from a single console. The data exchange is based on the Protocol Data Unit (PDU). There are 7 PDUs in the latest version of the SNMP protocol. Which of them sends a notification about the past event immediately, without waiting for the manager's request, and does not need confirmation of receipt?

- a. GetRequest
- b. Trap**
- c. GetNextRequest
- d. InformRequest

Question 85

Identify an adaptive SQL Injection testing technique by the description:

A testing technique is used to discover coding errors by inputting massive amounts of random data and observing the changes in the output.

- a. Functional Testing
- b. Dynamic Testing
- c. Static application security testing
- d. Fuzz Testing**

Question 86

Which of the following AI techniques is commonly used in intrusion detection systems (IDS) to identify anomalous network traffic?

- a. **Machine Learning**
- b. Expert Systems
- c. Computer Vision
- d. Natural Language Processing

Question 87

ISAPI filters is a powerful tool that is used to extend the functionality of IIS. However, improper use can cause huge harm. Why do EC-Council experts recommend that security analysts monitor the disabling of unused ISAPI filters?

- a. To prevent memory leaks
- b. To defend against wireless attacks
- c. To prevent leaks of confidential data
- d. **To defend against webserver attacks**

Question 88

An attacker stole financial information from a bank by compromising only a single server. After that, the bank decided to hire a third-party organization to conduct a full security assessment. Cybersecurity specialists have been provided with information about this case, and they need to provide an initial recommendation. Which of the following will be the best recommendation?

- a. Require all employees to change their passwords immediately
- b. Move the financial data to another server on the same IP subnet
- c. Issue new certificates to the web servers from the root certificate authority
- d. **Place a front-end web server in a demilitarized zone that only handles external web traffic**

Question 89

Your organization has adopted an AI-based malware detection system that leverages machine learning to detect malware based on behavioral analysis rather than traditional signature-based methods. Recently, the system flagged a new file that has no known

malware signature but behaves suspiciously by attempting to communicate with external servers. How does the AI likely determine this file as a potential threat?

- a. By comparing the file name with a database of known threats
- b. By scanning the file for encrypted code
- c. **By monitoring the file's network communication and behavior**
- d. By analyzing the file's hash against known malware

Question 90

What role does AI play in identifying zero-day vulnerabilities?

- a. AI ignores zero-day vulnerabilities
- b. **AI uses behavioral analytics to predict and uncover unknown vulnerabilities**
- c. AI can reverse-engineer patches
- d. AI can automatically deploy fixes

Question 91

When choosing a biometric system for your company, you should take into account the factors of system performance and whether they are suitable for you or not. What determines such a factor as the throughput rate?

- a. The probability that the system fails to detect a biometric input when presented correctly.
- b. The maximum number of sets of data that can be stored in the system
- c. The probability that the system incorrectly matches the input pattern to a non-matching template in the database
- d. **The data collection speeds, data processing speed, or enrolment time**

Question 92

Identify the algorithm according to the following description:

That wireless security algorithm was rendered useless by capturing packets and discovering the passkey in seconds. This vulnerability was strongly affected to TJ Maxx company. This vulnerability led to a network invasion of the company and data theft through a technique known as wardriving.

- a. **Wired Equivalent Privacy (WEP)**
- b. Wi-Fi Protected Access (WPA)
- c. Wi-Fi Protected Access 2 (WPA2)
- d. Temporal Key Integrity Protocol (TKIP)

Question 93

Which of the following documents describes the specifics of the testing, the associated violations and essentially protects both the organization's interest and third-party penetration tester?

- a. Project Scope
- b. Non-Disclosure Agreement
- c. **Rules of Engagement**
- d. Service Level Agreement

Question 94

Which of the following is NOT a benefit of AI in cybersecurity?

- a. **Completely eliminating human involvement in cybersecurity operations**
- b. Improved response times to incidents
- c. Faster detection of zero-day vulnerabilities
- d. Predictive threat analytics

Question 95

What is the primary goal of ethical hacking in the context of AI?

- a. To promote the ethical use of AI
- b. **To improve the security of AI systems**
- c. To exploit vulnerabilities in AI systems
- d. To develop new AI-based attack techniques

Question 96

Which of the following method of password cracking takes the most time?

- a. Dictionary attack
- b. Shoulder surfing
- c. Rainbow tables
- d. **Brute force**

Question 97

Which of the following stops vehicles from crashing through the doors of a building?

- a. Turnstile
- b. Traffic barrier
- c. Mantrap
- d. Bollards

Question 98

Ivan, a black-hat hacker, performs a man-in-the-middle attack. To do this, it uses a rogue wireless AP and embeds a malicious applet in all HTTP connections. When the victims went to any web page, the applet ran. Which of the following tools could Ivan probably use to inject HTML code?

- a. tcpdump
- b. Aircrack-ng
- c. **Ettercap**
- d. Wireshark

Question 99

How can AI be used to improve the effectiveness of phishing attacks?

- a. **By analyzing user behavior to identify vulnerable targets**
- b. By automating the delivery of phishing emails
- c. By bypassing spam filters
- d. By generating more convincing phishing emails

Question 100

Which of the following services run on TCP port 123 by default?

- a. DNS
- b. Telnet
- c. **NTP**
- d. POP3

Question 101

Confidential information is stored and processed on your company's servers, however, auditing has never been enabled. What of the following should be done before enabling the audit feature?

- a. Perform a cost/benefit analysis of the audit feature
- b. Perform a vulnerability scan of the system
- c. Allocate funds for staffing of audit log review
- d. **Determine the impact of enabling the audit feature**

Question 102

In what type of attack does the attacker forge the sender's IP address to gain access to protected systems and confidential data?

- a. IP fragmentation attack
- b. Source Routing
- c. IP forwarding
- d. **IP Spoofing**

Question 103

The company secretly hired hacker Ivan to attack its competitors before a major tender. Ivan did not start with complex technological attacks but decided to hit the employees and their reputation. To do this, he collected personal information about key employees of a competitor company. Then he began to distribute it in the open form on the Internet by adding false information about past racist statements of employees. As a result of the scandal in social networks and the censure of employees, competitors lost the opportunity to win the tender, and Ivan's work was done. What is the name of this form of attack?

- a. Daisy-chaining
- b. Piggybacking

- c. Vishing
- d. **Doxing**

Question 104

Identify a component of a risk assessment?

- a. **Administrative safeguards**
- b. Logical interface
- c. DMZ
- d. Physical security

Question 105

NIST defines risk management as the process of identifying, assessing, and controlling threats to an organization's capital and earnings. But what is the "risk" itself?

- a. An occurrence that actually or potentially jeopardizes the confidentiality, integrity, or availability of an information system
- b. The unauthorized disclosure, modification, or use of sensitive data
- c. **Potential that a threat will exploit vulnerabilities of an asset or group of assets**
- d. Weakness in an information system, system security procedures, internal controls, or implementation that could be exploited or triggered by a threat source

Question 106

The network elements of the telecom operator are located in the data center under the protection of firewalls and intrusion prevention systems. Which of the following is true for additional security measures?

- a. No additional measures are required since attacks and downtime are inevitable, and a backup site is required.
- b. No additional measures are required, since the attacker does not have physical access to the data center equipment
- c. Firewalls and intrusion detection systems are sufficient to ensure complete security
- d. **Periodic security checks and audits are required. Access to network elements should be provided by user IDs with strong passwords**

Question 107

Identify the structure designed to verify and authenticate the identity of individuals within the enterprise taking part in a data exchange?

- a. SOA
- b. Biometrics
- c. **PKI**
- d. single sign-on

Question 108

The attacker managed to gain access to Shellshock, and now he can execute arbitrary commands and gain unauthorized access to many Internet-facing services. Which of the following operating system can't be affected by an attacker yet?

- a. OS X
- b. Unix
- c. **Windows**
- d. Linux

Question 109

One of the most popular tools in the pentester's arsenal - John the Ripper is designed for...

- a. Automation of the process of detecting and exploiting the SQL injection vulnerability
- b. **Test password strength, brute-force encrypted or hashed passwords, and crack passwords via dictionary attacks**
- c. Search for various default and insecure files, configurations, and programs on any type of web servers
- d. Discover hosts and services on a computer network by sending packets and analyzing the responses

Question 110

Identify a low-tech way of gaining unauthorized access to information?

- a. Scanning
- b. Sniffing

- c. Eavesdropping
- d. **Social engineering**

Question 111

When getting information about the web server, you should be familiar with methods GET, POST, HEAD, PUT, DELETE, TRACE. There are two critical methods in this list: PUT (upload a file to the server) and DELETE (delete a file from the server). When using nmap, you can detect all these methods. Which of the following nmap scripts will help you detect these methods?

- a. **http-methods**
- b. http-headers
- c. http ETag
- d. http enum

Question 112

Enumeration is a process which establishes an active connection to the target hosts to discover potential attack vectors in the system, and the same can be used for further exploitation of the system. What type of enumeration is used to get shared resources on individual hosts on the network and a list of computers belonging to the domain?

- a. **Netbios enumeration**
- b. NTP enumeration
- c. SNMP enumeration
- d. SMTP enumeration

Question 113

There are different ways of pentest of a system, network, or application in information security based on how much information you have about the target. There's black box testing, white box testing, and gray box testing. Which of the statements is true about grey-box testing?

- a. The tester has full access to the internal structure
- b. The tester does not have access at all
- c. The tester is unaware of the internal structure

- d. **The tester only partially knows the internal structure**

Question 114

Alex, an employee of a law firm, receives an email with an attachment "Court_Notice_09082020.zip". There is a file inside the archive "Court_Notice_09082020.zip.exe". Alex does not notice that this is an executable file and runs it. After that, a window appears with the notification "This word document is corrupt" and at the same time, malware copies data to APPDATA\local directory takes place in the background and begins to beacon to a C2 server to download additional malicious binaries. What type of malware has Alex encountered?

- a. Worm
- b. Key-Logger
- c. **Trojan**
- d. Macro Virus

Question 115

Gabriella uses Google search operators, which allow you to optimize and expand the capabilities of regular search. What will be the result of this request: *site:eccouncil.org discount -ilearn* ?

- a. Results from the ec-council website except for discounts and the ilearn format
- b. The results that match the entire query
- c. **Results about all discounts from the site eccouncil.org except for the ilearn format**
- d. Results about all discounts from the site ec-council.org for the ilearn training format

Question 116

Shortly after replacing the outdated equipment, John, the company's system administrator, discovered a leak of critical customer information. Moreover, among the stolen data was the new user's information that excludes incorrect disposal of old equipment. IDS did not notice the intrusion, and the logging system shows that valid credentials were used. Which of the following is most likely the cause of this problem?

- a. Industrial Espionage

- b. Zero-day vulnerabilities
- c. **Default Credential**
- d. NSA backdoor

Question 117

What is the minimum number of network connections needed for a multi-homed firewall?

- a. 4
- b. **2**
- c. 5
- d. 8

Question 118

In which phase of the ethical hacking process can Google hacking be used?

For example: allintitle: root passwd

- a. Maintaining Access
- b. Gaining Access
- c. Scanning and Enumeration
- d. **Reconnaissance**

Question 119

Which of the following is a common IDS evasion technique?

- a. **Unicode characters**
- b. Subnetting
- c. Spyware
- d. Port knocking

Question 120

The company is trying to prevent the security breach by applying a security policy in which all Web browsers must automatically delete their HTTP browser cookies upon termination. Identify the security breach that the company is trying to prevent?

- a. **Attempts by attackers to access websites that trust the Web browser user by stealing the employee's authentication credentials**
- b. Attempts by attackers to determine the employee's web browser usage patterns
- c. Attempts by attackers to access the user and password information stored in the company's SQL database
- d. Attempts by attackers to access passwords stored on the employee's computer

Question 121

The SOC analyst of the company wants to track the transfer of files over the unencrypted FTP protocol, which filter for the Wireshark sniffer should he use?

- a. tcp.port = 23
- b. **tcp.port == 21**
- c. tcp.port == 80
- d. tcp.port == 443

Question 122

Identify which of the following will provide you with the most information about the system's security posture?

- a. **Port scanning, banner grabbing, service identification**
- b. Wardriving, warchalking, social engineering
- c. Social engineering, company site browsing, tailgating
- d. Phishing, spamming, sending trojans

Question 123

Identify a security policy that defines using of a VPN for gaining access to an internal corporate network?

- a. Information protection policy
- b. Access control policy
- c. **Remote access policy**
- d. Network security policy

Question 124

Shellshock is a serious bug in the Bash command-line interface shell that allows an attacker to execute commands by gaining unauthorized access to computer systems.

```
env x=`(){{:}};echo exploit` bash -c 'cat /etc/passwd'
```

What is the result of executing this query on a vulnerable host?

- a. Deleting the passwd file
- b. **Display of the contents of the passwd file**
- c. Copying the contents of the passwd file
- d. Creating a passwd file

Question 125

What is the name of the practice of collecting information from published or otherwise publicly available sources?

- a. **Open-source intelligence**
- b. Artificial intelligence
- c. Human intelligence
- d. Social intelligence